The Standard Softmax logic from the flash attention paper

$$S = QK^T, \quad P = \text{softmax}(S), \quad O = PV$$

Let's take $O = PV$

In topics like thermodynamics (for different thermodynamics processes) or let's say wave function we do partial differentiation but here dO is the output of matrix multiplication between P and value V.

$$dO = P\,dV + dP\,V$$

In neural networks we care about $L(O)$ because it tells us how to change the model's parameters to reduce the loss.

Let us consider a flow state:

$$\text{Input} \to \text{Layer A} \to \text{Layer B} \to \text{Loss } L$$

When we are trying to compute the gradient for Layer A, the gradient coming from B is called the upstream gradient.

For a scalar function $L(O)$

> **Note:** : meaning inner product basically multiply each element and then sum.

$$dL = \sum_{i,j} (L_O)_{ij}\, dO_{ij}$$

$$dL = L_O : dO$$

The upstream gradient $G = L_O$

$$dL = G : (P\,dV + dP\,V)$$

$$dL = G : (P\,dV) + G : (dP\,V)$$

Lets write the above term in index notation again:

$$G : (P\,dV) = \sum_{i,j} G_{ij}(P\,dV)_{ij} \quad \text{— i)}$$

$$G : (dP\,V) = \sum_{i,j} G_{ij}(dP\,V)_{ij} \quad \text{— ii)}$$

Lets understand why that transpose came into place by proving one of the equations.

From equation i)

$$G : (P\,dV) = \sum_{i,j} G_{ij}(P\,dV)_{ij}$$

Let us only take the term $P\,dV$ (Note:- this is not thermodynamics or waive function the output is defined as the result of something so we cannot keep something constant)

$$(P\,dV)_{ij} = \sum_{k} P_{ik}(dV)_{kj} \quad \text{— iii)}$$

> **Note:** From the attention score formula P which is the output of the softmax and then V which is the value is matrix multiplied.

Now multiply that term $(P\,dV)_{ij}$ with upstream gradient G

From equation iii)

$P_{ik}(dV)_{kj}$ and G_{ij} will have an inner product.

$$G : (P\,dV) = \sum_{i,j,k} G_{ij} P_{ik}(dV)_{kj}$$

We are now regrouping by multiplying with $(dV)_{kj}$

$$\sum_{k,j}\left(\sum_i P_{ik}G_{ij}\right)(dV)_{kj} \quad \text{— iv)}$$

Basically you can group this with the involved terms.

Now by the definition of transpose from school level math.

$$(P^T)_{ki} = P_{ik}$$

Common point of confusing here value of the index are same:

$$\sum_{k,j}\left(\sum_i (P^T)_{ki}G_{ij}\right)(dV)_{kj} \quad \text{from eq iv)}$$

$$G : (P\,dV) = (P^T G) : dV$$

Reading off the gradient for the first and second part of the above equation.

$$dL = (P^T G) : dV$$

$$L_V = P^T L_O \quad \text{— a)}$$

$$dL = (GV^T) : dP$$

$$L_P = L_O V^T \quad \text{— b)}$$

Let's unfold exactly what is going on at the matrix level.

The Value matrix has shape (3×3):

$$V = \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix}$$

2. Attention Probability Matrix (P)

Each row of P is the result of applying the `softmax` function across the corresponding row of the attention score matrix S.

Let the row-wise denominator sums be defined as:

- Row 1: $\sum e^{s_1} = e^{s_{11}} + e^{s_{12}} + e^{s_{13}}$
- Row 2: $\sum e^{s_2} = e^{s_{21}} + e^{s_{22}} + e^{s_{23}}$
- Row 3: $\sum e^{s_3} = e^{s_{31}} + e^{s_{32}} + e^{s_{33}}$

$$P = \text{softmax}(S) = \begin{bmatrix} \frac{e^{s_{11}}}{\sum e^{s_1}} & \frac{e^{s_{12}}}{\sum e^{s_1}} & \frac{e^{s_{13}}}{\sum e^{s_1}} \\ \frac{e^{s_{21}}}{\sum e^{s_2}} & \frac{e^{s_{22}}}{\sum e^{s_2}} & \frac{e^{s_{23}}}{\sum e^{s_2}} \\ \frac{e^{s_{31}}}{\sum e^{s_3}} & \frac{e^{s_{32}}}{\sum e^{s_3}} & \frac{e^{s_{33}}}{\sum e^{s_3}} \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix}$$

3. Output Matrix ($O = PV$)

Multiplying rows of P by columns of V gives us the fully expanded elements of the output matrix O:

$$O = \begin{bmatrix} p_{11}v_{11} + p_{12}v_{21} + p_{13}v_{31} & p_{11}v_{12} + p_{12}v_{22} + p_{13}v_{32} & p_{11}v_{13} + p_{12}v_{23} + p_{13}v_{33} \\ p_{21}v_{11} + p_{22}v_{21} + p_{23}v_{31} & p_{21}v_{12} + p_{22}v_{22} + p_{23}v_{32} & p_{21}v_{13} + p_{22}v_{23} + p_{23}v_{33} \\ p_{31}v_{11} + p_{32}v_{21} + p_{33}v_{31} & p_{31}v_{12} + p_{32}v_{22} + p_{33}v_{32} & p_{31}v_{13} + p_{32}v_{23} + p_{33}v_{33} \end{bmatrix}$$

Which maps directly to the standard layout:

$$O = \begin{bmatrix} O_{11} & O_{12} & O_{13} \\ O_{21} & O_{22} & O_{23} \\ O_{31} & O_{32} & O_{33} \end{bmatrix}$$

The upstream gradient for $P = \mathrm{softmax}(S) = \frac{\partial L}{\partial P}$

Quotient Rule:

The quotient rule for derivatives states:

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

where u and v are functions of x, and u' and v' are their respective derivatives.

Now, what we want to do is evaluate this row-wise for the attention probability matrix P.

Let's first look closely at the numerator and the denominator of the softmax fraction.

For example, consider the position p_{11}:

- **Numerator (N):** $N = e^{s_{11}}$
- **Denominator (D):** $D = e^{s_{11}} + e^{s_{12}} + e^{s_{13}}$

Let's take the partial derivative with respect to the input score s_{11}:

- The derivative of the numerator with respect to s_{11} is:

$$N' = \frac{\partial N}{\partial s_{11}} = e^{s_{11}}$$

- The derivative of the denominator with respect to s_{11} is:

$$D' = \frac{\partial D}{\partial s_{11}} = e^{s_{11}}$$

The Full Chain Rule across the Row

Here we apply the chain rule to compute the gradient with respect to the input logit s_{11}.

Because the denominator is a shared sum across the entire row, changing s_{11} alters the value of *every single probability element* in that row (p_{11}, p_{12}, and p_{13}). This row-wise coupling is exactly why parallel reduction in a GPU kernel requires careful shared memory communication or warp shuffles.

Using the total derivative chain rule, we must sum the gradient contributions flowing back through all elements in that row:

In this case, we have $\frac{\partial L}{\partial P}$ as our **upstream gradient** and $\frac{\partial P}{\partial S_{11}}$ as our **local gradient**. Therefore,

$$\frac{\partial L}{\partial S_{11}} = \frac{\partial L}{\partial P} \cdot \frac{\partial P}{\partial S_{11}}.$$

This follows directly from the chain rule.

$$\frac{\partial L}{\partial s_{11}} = \frac{\partial L}{\partial p_{11}}\frac{\partial p_{11}}{\partial s_{11}} + \frac{\partial L}{\partial p_{12}}\frac{\partial p_{12}}{\partial s_{11}} + \frac{\partial L}{\partial p_{13}}\frac{\partial p_{13}}{\partial s_{11}}$$

In a more compact notation using the upstream gradient components $(L_P)_{1j}$:

$$\frac{\partial L}{\partial s_{11}} = \sum_{j=1}^{3} \frac{\partial L}{\partial p_{1j}}\frac{\partial p_{1j}}{\partial s_{11}}$$

$$\frac{\partial L}{\partial s_{11}} = \left(\frac{\partial L}{\partial p_{11}} \cdot \frac{\partial p_{11}}{\partial s_{11}}\right) + \left(\frac{\partial L}{\partial p_{12}} \cdot \frac{\partial p_{12}}{\partial s_{11}}\right) + \left(\frac{\partial L}{\partial p_{13}} \cdot \frac{\partial p_{13}}{\partial s_{11}}\right) \quad \text{— (Equation V)}$$

Now in our example, at the position p_{11}:

$$p_{11} = \frac{e^{s_{11}}}{e^{s_{11}} + e^{s_{12}} + e^{s_{13}}}$$

Now taking the partial derivative of p_{11} with respect to s_{11} using the quotient rule:

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{e^{s_{11}}\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right) - e^{s_{11}}\left(e^{s_{11}}\right)}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{e^{s_{11}}e^{s_{11}} + e^{s_{11}}e^{s_{12}} + e^{s_{11}}e^{s_{13}} - e^{s_{11}}e^{s_{11}}}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{e^{s_{11}}e^{s_{12}} + e^{s_{11}}e^{s_{13}}}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{\left(e^{s_{11}}e^{s_{11}} + e^{s_{11}}e^{s_{12}} + e^{s_{11}}e^{s_{13}}\right) - e^{s_{11}}e^{s_{11}}}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

Now, factor out $e^{s_{11}}$ from the first group in the numerator:

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{e^{s_{11}}\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right) - \left(e^{s_{11}}\right)^2}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

Next, split the fraction into two separate terms across the minus sign:

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{e^{s_{11}}\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2} - \frac{\left(e^{s_{11}}\right)^2}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

Cancel out the shared terms in the first fraction:

$$\frac{\partial p_{11}}{\partial s_{11}} = \frac{e^{s_{11}}}{e^{s_{11}} + e^{s_{12}} + e^{s_{13}}} - \left(\frac{e^{s_{11}}}{e^{s_{11}} + e^{s_{12}} + e^{s_{13}}}\right)^2$$

Since $p_{11} = \frac{e^{s_{11}}}{e^{s_{11}} + e^{s_{12}} + e^{s_{13}}}$, we can substitute it directly back into the terms:

$$\frac{\partial p_{11}}{\partial s_{11}} = p_{11} - p_{11}^2$$

There's nothing to be scared of. Think of this as multiplying by the upstream gradient; it's just the quotient rule applied to the derivative term and then rearranging the terms.

For position p_{12}, because this is a partial derivative, everything else is a constant.

$$\frac{\partial p_{12}}{\partial s_{11}} = \frac{0 \cdot \left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right) - e^{s_{12}}e^{s_{11}}}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

$$\frac{\partial p_{12}}{\partial s_{11}} = \frac{-e^{s_{11}}e^{s_{12}}}{\left(e^{s_{11}} + e^{s_{12}} + e^{s_{13}}\right)^2}$$

$$= -p_{11}p_{12}$$

In a similar way for position p_{13}:

$$\frac{\partial p_{13}}{\partial s_{11}} = -p_{11}p_{13}$$

Remember we are doing a row wise opreation here.

From Equation V:

$$\frac{\partial L}{\partial s_{11}} = \frac{\partial L}{\partial p_{11}}(p_{11} - p_{11}^2) + \frac{\partial L}{\partial p_{12}}(-p_{11}p_{12}) + \frac{\partial L}{\partial p_{13}}(-p_{11}p_{13})$$

In a similar way we can calculate for row 2 and row 3.

For s_{12}, we take:

$$\frac{\partial L}{\partial s_{12}} = \left(\frac{\partial L}{\partial p_{11}} \cdot (-p_{11}p_{12})\right) + \left(\frac{\partial L}{\partial p_{12}} \cdot p_{12}(1-p_{12})\right) + \left(\frac{\partial L}{\partial p_{13}} \cdot (-p_{12}p_{13})\right)$$

For s_{13}, we take:

$$\frac{\partial L}{\partial s_{13}} = \left(\frac{\partial L}{\partial p_{11}} \cdot (-p_{11}p_{13})\right) + \left(\frac{\partial L}{\partial p_{12}} \cdot (-p_{12}p_{13})\right) + \left(\frac{\partial L}{\partial p_{13}} \cdot p_{13}(1-p_{13})\right)$$

Now lets take a look at the original matrix.

$$P = \text{softmax}(S) = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix}$$

Lets pull out gradient vector from equation vi meaning from the first row for example

$$\frac{\partial L}{\partial s_{1:}} = \begin{bmatrix} \frac{\partial L}{\partial s_{11}} & \frac{\partial L}{\partial s_{12}} & \frac{\partial L}{\partial s_{13}} \end{bmatrix}$$

For P which is the output of the softmax

$$\frac{\partial L}{\partial p_{1:}} = \begin{bmatrix} \frac{\partial L}{\partial p_{11}} & \frac{\partial L}{\partial p_{12}} & \frac{\partial L}{\partial p_{13}} \end{bmatrix}$$

will we wrap those vectors like this, if you multiply then you get the same result as above and remember all this for a single row.

$$\frac{\partial L}{\partial S_1} = \frac{\partial L}{\partial P_1} \cdot \begin{bmatrix} p_{11}(1-p_{11}) & -p_{11}p_{12} & -p_{11}p_{13} \\ -p_{11}p_{12} & p_{12}(1-p_{12}) & -p_{12}p_{13} \\ -p_{11}p_{13} & -p_{12}p_{13} & p_{13}(1-p_{13}) \end{bmatrix}$$

Now we will split this into Jacobian matrix and if you have doubt then multiply and see a small example its the same.

$$J(P_1) = \text{diag}(P_1) - P_1^T P_1$$

Where:

$$\text{diag}(P_1) = \begin{bmatrix} p_{11} & 0 & 0 \\ 0 & p_{12} & 0 \\ 0 & 0 & p_{13} \end{bmatrix}$$

$$P_1^T P_1 = \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} & p_{13} \end{bmatrix} = \begin{bmatrix} p_{11}^2 & p_{11}p_{12} & p_{11}p_{13} \\ p_{11}p_{12} & p_{12}^2 & p_{12}p_{13} \\ p_{11}p_{13} & p_{12}p_{13} & p_{13}^2 \end{bmatrix}$$

$$J(P_1) = \begin{bmatrix} p_{11} & 0 & 0 \\ 0 & p_{12} & 0 \\ 0 & 0 & p_{13} \end{bmatrix} - \begin{bmatrix} p_{11}^2 & p_{11}p_{12} & p_{11}p_{13} \\ p_{11}p_{12} & p_{12}^2 & p_{12}p_{13} \\ p_{11}p_{13} & p_{12}p_{13} & p_{13}^2 \end{bmatrix} = \begin{bmatrix} p_{11}(1-p_{11}) & -p_{11}p_{12} & -p_{11}p_{13} \\ -p_{11}p_{12} & p_{12}(1-p_{12}) & -p_{12}p_{13} \\ -p_{11}p_{13} & -p_{12}p_{13} & p_{13}(1-p_{13}) \end{bmatrix}$$

And this is for one row. This part is to grab the concept right and we will work on writing kernels which will make this procress even transparent.

Now we will be looking at:

$$S = QK^T$$

Here, the upstream gradient backpropagating into this operation will be $\frac{\partial L}{\partial S}$.

The dot product for an individual element s_{ij} is given by:

$$s_{ij} = \sum_k q_{ik} K_{jk}$$

Now let's isolate the component q_{ij} to find its gradient. Using the chain rule:

$$\frac{\partial L}{\partial q_{ij}} = \sum_k \frac{\partial L}{\partial s_{ik}} \frac{\partial s_{ik}}{\partial q_{ij}}$$

Therefore, the local gradient is:

$$\frac{\partial s_{ik}}{\partial q_{ij}} = K_{jk}$$

Substituting this back into the chain rule expression yields the matrix multiplication form:

$$\frac{\partial L}{\partial Q} = \frac{\partial L}{\partial S} K$$

I will derive O = PV in better way inorder for me to understand better, this is for me to look back at if I forget how does this work and implemnting in code will even make it clear.

$$V = \begin{bmatrix} V_{11} & V_{12} & V_{13} \\ V_{21} & V_{22} & V_{23} \\ V_{31} & V_{32} & V_{33} \end{bmatrix}$$

$$O = \begin{bmatrix} O_{11} & O_{12} & O_{13} \\ O_{21} & O_{22} & O_{23} \\ O_{31} & O_{32} & O_{33} \end{bmatrix}$$

$$K = \begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix}$$

O = PV

Here, the upstream gradient backpropagating from the loss function down to this layer is:

$$\frac{\partial L}{\partial O} = \begin{bmatrix} \frac{\partial L}{\partial O_{11}} & \frac{\partial L}{\partial O_{12}} & \frac{\partial L}{\partial O_{13}} \\ \frac{\partial L}{\partial O_{21}} & \frac{\partial L}{\partial O_{22}} & \frac{\partial L}{\partial O_{23}} \\ \frac{\partial L}{\partial O_{31}} & \frac{\partial L}{\partial O_{32}} & \frac{\partial L}{\partial O_{33}} \end{bmatrix}$$

$$O = \begin{bmatrix} O_{11} & O_{12} & O_{13} \\ O_{21} & O_{22} & O_{23} \\ O_{31} & O_{32} & O_{33} \end{bmatrix} = \begin{bmatrix} p_{11}v_{11} + p_{12}v_{21} + p_{13}v_{31} & p_{11}v_{12} + p_{12}v_{22} + p_{13}v_{32} & p_{11}v_{13} + p_{12}v_{23} + p_{13}v_{33} \\ p_{21}v_{11} + p_{22}v_{21} + p_{23}v_{31} & p_{21}v_{12} + p_{22}v_{22} + p_{23}v_{32} & p_{21}v_{13} + p_{22}v_{23} + p_{23}v_{33} \\ p_{31}v_{11} + p_{32}v_{21} + p_{33}v_{31} & p_{31}v_{12} + p_{32}v_{22} + p_{33}v_{32} & p_{31}v_{13} + p_{32}v_{23} + p_{33}v_{33} \end{bmatrix}$$

$$O_{11} = p_{11}v_{11} + p_{12}v_{21} + p_{13}v_{31}$$

This opreation happens column wise.

From our matrix equation for O, the terms containing v_{11} appear in the first column (O_{11}, O_{21}, O_{31}):

$$O_{11} = p_{11}v_{11} + p_{12}v_{21} + p_{13}v_{31}$$

$$O_{21} = p_{21}v_{11} + p_{22}v_{21} + p_{23}v_{31}$$

$$O_{31} = p_{31}v_{11} + p_{32}v_{21} + p_{33}v_{31}$$

Applying the multivariate chain rule:

$$\frac{\partial L}{\partial v_{11}} = \frac{\partial L}{\partial O_{11}} \frac{\partial O_{11}}{\partial v_{11}} + \frac{\partial L}{\partial O_{21}} \frac{\partial O_{21}}{\partial v_{11}} + \frac{\partial L}{\partial O_{31}} \frac{\partial O_{31}}{\partial v_{11}}$$

Lets take the expression

$$O_{11} = p_{11}v_{11} + p_{12}v_{21} + p_{13}v_{31}$$

$$\frac{\partial O_{11}}{\partial v_{11}} = p_{11} \frac{\partial v_{11}}{\partial v_{11}} = p_{11}$$

Following the same logic for the other elements in that column:

$$\frac{\partial O_{21}}{\partial v_{11}} = p_{21}$$

$$\frac{\partial O_{31}}{\partial v_{11}} = p_{31}$$

Substituting those local derivatives back into the chain rule expression gives:

$$\frac{\partial L}{\partial v_{11}} = \frac{\partial L}{\partial O_{11}} p_{11} + \frac{\partial L}{\partial O_{21}} p_{21} + \frac{\partial L}{\partial O_{31}} p_{31}$$

So this is equal to the:-

$$\frac{\partial L}{\partial V} = P^T G$$

FlashAttention Backpropagation — Derivation and Implementation

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References

- Dao et al., FlashAttention (2022) https://arxiv.org/abs/2205.14135